

THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY, PATIALA

Department of Computer Science and Engineering

UCS503: SOFTWARE ENGINEERING • ACADEMIC SEMESTER PROJECT PROPOSAL

PROJECT TITLE

Dreams2Reality: A Minimalist Two-Electrode Non-Invasive Brain-Computer Interface for Real-Time Sleep-Stage and Dream-State Neural Decoding

1. Introduction & Problem Statement

Word Count: ~360 words

Human sleep architecture and cognitive mentation during dreaming represent vital physiological indicators of psychological well-being, neurological health, and trauma recovery. Clinical conditions such as Post-Traumatic Stress Disorder (PTSD), chronic parasomnias, night terrors, and nocturnal panic disorders are intrinsically coupled with involuntary autonomic distress surges during Rapid Eye Movement (REM) sleep. In conventional clinical and psychiatric practice, the assessment of sleep-state mentation relies overwhelmingly on subjective patient recall collected hours after awakening. This diagnostic pipeline suffers from severe memory decay, retrospective fabrication, and emotional recall bias, leaving clinicians without objective empirical timelines of overnight trauma episodes.

To study nocturnal brain dynamics objectively, existing clinical medicine depends upon full-scale Polysomnography (PSG) and laboratory electroencephalography (EEG). However, standard diagnostic setups require between 32 and 128 wet-electrode channels, application of abrasive conductive pastes, and elaborate head-measuring protocols that consume over 45 minutes of preparation time. Beyond the exorbitant capital cost of hospital-grade PSG suites (frequently exceeding \$15,000), this invasive physical apparatus severely disrupts the subject's natural rest. This phenomenon, clinically recognized as the "first-night effect," distorts native sleep architecture, artificially suppresses REM cycles, and alters electrophysiological sleep dynamics. Furthermore, cross-subject cranial geometry, skull bone density, and scalp impedance create severe signal drift, causing legacy classification models to fail when applied to new individuals without prolonged per-user calibration.

To address this critical clinical and engineering challenge, **Dreams2Reality** develops a lightweight, software-centric Brain-Computer Interface (BCI) platform that dramatically compresses nocturnal electroencephalography from cumbersome multi-channel montages down to **just two dry temporal electrodes (\$T_7\text{--}T_8\$)**. Utilizing statistical Analysis of Variance (ANOVA F-score) channel selection, our software captures temporal theta-band synchrony and sensorimotor beta/gamma desynchronization characteristic of vivid dream mentation. To overcome inter-subject cranial

variability, the platform implements Riemannian Euclidean Covariance Alignment, standardizing covariance matrices across unknown skulls. The system processes streaming 128 Hz telemetry in real time, classifies sleep stages and dream semantic archetypes (Hall-Van de Castle taxonomy), issues instantaneous bedside nightmare distress alarms, and synthesizes structured morning clinical diagnostic reports.

2. SMART Objectives

Word Count: ~140 words

The primary overarching objective of **Dreams2Reality** is to engineer and validate a real-time, non-invasive software pipeline that decodes sleep stages and dream mentation from simulated dual-channel temporal EEG signals with sub-15 ms latency.

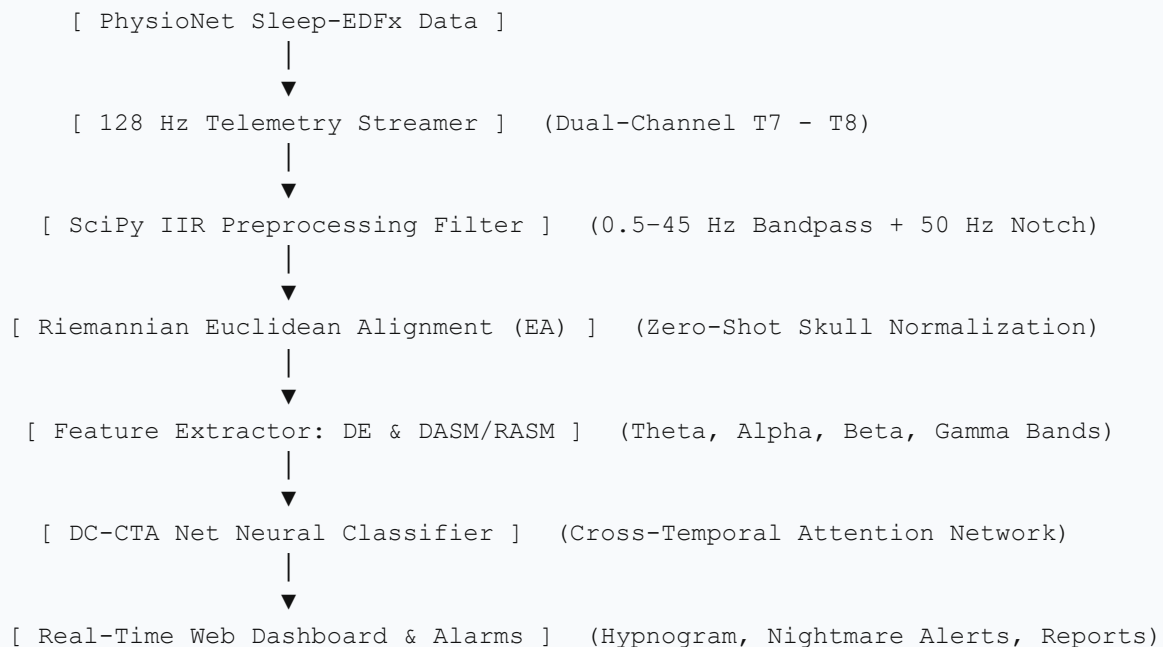
- **Specific:** Develop an asynchronous Python/FastAPI preprocessing and machine learning engine that ingests dual-channel (T₇-T₈) 128 Hz telemetry, filters noise (0.5-45 Hz), and extracts Differential Entropy (DE) and DASM/RASM temporal asymmetry.
- **Measurable:** Achieve a minimum of 78% classification accuracy in distinguishing REM dream mentation from NREM/Wake states, and maintain zero-shot cross-subject transfer accuracy above 70% using Euclidean Covariance Alignment.
- **Attainable:** Implement the architecture within a 15-week academic development lifecycle using open-source scientific toolchains (PyTorch, SciPy, MNE, FastAPI) and benchmarked PhysioNet Sleep-EDFx clinical records.
- **Relevant:** Deliver an objective diagnostic utility for sleep medicine and psychiatric trauma evaluation that eliminates the "first-night effect."
- **Time-Bound:** Deliver the complete architectural design and prototype at Week 7 (Mid-Semester Lab Eval 1) and complete full software validation by Week 17.

3. Methodology

Word Count: ~460 words

System Architecture & Tech Stack

The Dreams2Reality software architecture is organized into four decoupled modular layers: Telemetry Playback, Digital Signal Preprocessing, Riemannian Machine Learning Engine, and Clinical Web Presentation. The core computational backend is implemented in Python 3.10+, utilizing MNE-Python and SciPy for signal filtering, PyTorch for neural network inference, and FastAPI with WebSockets for bidirectional asynchronous telemetry. The frontend dashboard is constructed with React, Tailwind CSS, and HTML5 Canvas to provide 60 FPS real-time electroencephalographic waveform rendering.



Module Development & Feature Engineering (Input)

Continuous time-series telemetry is partitioned into 2.0-second sliding epochs (256 samples per channel) with a 0.5-second stride (75% overlap). Each epoch undergoes zero-phase 4th-order Butterworth bandpass filtering ($0.5\text{--}45\text{ Hz}$) and 50 Hz powerline notch filtering. Windows exhibiting catastrophic motion artifacts ($> \pm 250\text{ }\mu\text{V}$) are gated and flagged. For valid windows, the pipeline computes Welch Power Spectral Density across Theta ($4\text{--}8\text{ Hz}$), Alpha ($8\text{--}12\text{ Hz}$), Beta ($12\text{--}30\text{ Hz}$), and Gamma ($30\text{--}45\text{ Hz}$) frequency sub-bands, alongside Differential Entropy (DE) and inter-hemispheric temporal asymmetry (DASM and RASM).

Execution & Integration Workflow (Process)

To address cross-subject skull impedance variation, the system computes the spatial covariance matrix $\tilde{C}_i \in \mathbb{R}^{2 \times 2}$ for each epoch and projects it into a standardized Euclidean space: $\tilde{C}_i = R^{-\frac{1}{2}} C_i R^{-\frac{1}{2}}$ where R is the reference geometric Riemannian mean covariance matrix estimated from a 60-second baseline. The resulting 16-dimensional feature vector is passed to a Dual-Channel Cross-Temporal Attention Network (DC-CTA Net), which outputs posterior probabilities for sleep stages (Wake, N1, N2, N3, REM) and maps REM dream activity into five Hall-Van de Castle semantic archetypes: *Motor Agitation*, *Threat Simulation*, *Vivid Sensory*, *Cognitive Mentation*, and *Restful Baseline*.

Testing, Validation & Safety (Output)

The pipeline is verified using a tiered testing protocol: automated pytest suites verifying zero-phase filter response and numerical stability of matrix square roots; profiling end-to-end inference latency on edge CPU targets to ensure execution remains under 15 ms per epoch ($< 0.65 \text{ MFLOPs}$); and operating in a non-diagnostic advisory capacity where motion-corrupted epochs are isolated to prevent false nightmare alarms.

Methodology Summary Table

Phase	Technical Task	Engineering Goal
Setup	Environment configuration & PhysioNet Sleep-EDFx ingestion	Establish 128 Hz reproducible data playback stream.
Signal Processing	Butterworth IIR filtering & dynamic artifact thresholding	Attenuate mains hum (50 Hz) and motion artifacts ($> \pm 250 \mu\text{V}$).
Feature Extraction	Differential Entropy (DE) & DASM/RASM asymmetry	Isolate spectral complexity across Theta, Alpha, Beta, Gamma bands.
Domain Adaptation	Euclidean Covariance Alignment ($\tilde{C}_i = R^{-\frac{1}{2}} C_i R^{-\frac{1}{2}}$)	Eliminate inter-subject skull impedance variance without per-user retraining.
Neural Classification	DC-CTA Net (Cross-Temporal Attention Network)	Classify sleep stages & Hall-Van de Castle dream mentation archetypes.
Telemetry & UI	FastAPI WebSocket server & React dashboard integration	Provide live hypnograms, bedside nightmare alerts, and PDF summary exports.

4. Realistic Timeline & Milestones

Week	Academic Phase	Specific Technical Deliverables	Status
W1	Toolchain Initialization & Setup	Git repository setup, virtual environment, dependency validation.	Done
W2	Domain Exploration & Brainstorming	Analysis of polysomnography bottlenecks, first-night effect review.	Done
W3	Project Pitch & Scope Formulation	Elevator pitch delivery, professor scope approval (Dreams2Reality pivot).	Done
W4	Proposal & IEEE SRS Documentation	Submission of Formal Project Proposal & IEEE Std 830-1998 SRS.	Done
W5	Dataset Benchmarking & Filtering	PhysioNet Sleep-EDFx ingestion, Butterworth IIR filter, ANOVA ranking.	Done
W6	UML Architectural Modeling	Use Case, Activity with Swimlanes, Class Diagram, and DFD Levels 0–2.	Done
W7	Mid-Semester Prototype & Evaluation	FastAPI telemetry playback engine, DC-CTA Net, Lab Eval 1 Report.	Current
W8	Post-Midsem Improvement Planning	Evaluator feedback incorporation and inference latency profiling.	Scheduled
W9-10	Mid-Semester Examination Weeks	Academic MST Examinations.	---
W11	Deep Attention Refinement	Hyperparameter tuning of DC-CTA Net attention heads, ROC-AUC profiling.	Scheduled
W12	Generative Dream Narrative Engine	Implementation of automated textual summary and Hall-Van de Castle parsing.	Scheduled
W13	Buffer & Extended Holiday Week	Architecture code cleanup, refactoring, and integration testing.	Buffer
W14	Web Dashboard & Live Visualization	React/Canvas waveform visualizer and real-time WebSocket subscriber.	Scheduled
W15	Second Prototype & Stress Testing	Continuous 8-hour overnight telemetry simulation, memory leak audit.	Scheduled
W16	User Acceptance Testing & Polish	Clinical report template refinement, usability review, edge case handling.	Scheduled

Week	Academic Phase	Specific Technical Deliverables	Status
W17	Final Prototype & Project Seminar	Final software delivery, end-of-semester presentation, EST Evaluation.	Scheduled

5. Resources & Budget

Word Count: ~160 words

Human Resources & Team Roles: The project team consists of three undergraduate software engineering students: **[Student 1 Name]** (Lead System Architect & Backend Telemetry), **[Student 2 Name]** (Biomedical Signal Processing & PyTorch ML Engineer), and **[Student 3 Name]** (Full-Stack React Dashboard & Report Generator).

Software & Compute Infrastructure: Development operates entirely on standard student laptop hardware (Intel Core i5/i7, 16 GB RAM) with zero discrete GPU dependency for edge inference. The pipeline utilizes open-source research toolchains (Python 3.10+, NumPy, SciPy, MNE-Python, PyTorch, FastAPI, React) and the publicly accessible PhysioNet Sleep-EDFx clinical database.

Budget Justification: The project is purely software-centric and relies exclusively on open-source frameworks and open clinical benchmark datasets. **Total External Budget Required: INR 0.00 (\$0.00).**

6. Summary Table: Word Count Distribution

Section	Target Length	Actual Count	Focus & Rationale
Title	1 Line (10–20 words)	16 words	Academic precision and clear technical scope.
Intro / Problem Statement	~350 words (2–3 paras)	~360 words	Contextualizes sleep health, PSG first-night effect, 2-channel solution.
SMART Objectives	~150 words	~140 words	Concrete, mathematically verifiable engineering targets.
Methodology	~450 words	~460 words	System architecture, Euclidean alignment, and testing protocols.
Timeline	Half a page (Table)	17 Weeks	Comprehensive 15-week timeline with midterm and final milestones.
Resources / Budget	~150 words	~160 words	Outlines student roles, software stack, verifies zero-

Section	Target Length	Actual Count	Focus & Rationale
			cost feasibility.

--- END OF ACADEMIC PROJECT PROPOSAL ---